Aurora Stays Q&A

Retrieval-Augmented Generation (RAG) POC

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Executive Summary

- Built a source-grounded RAG assistant for Aurora Stays hotel-policy questions.
- Four policy PDFs are converted to structured Markdown, chunked, embedded, and indexed in FAISS.
- Selected baseline: k=5, MMR lambda=0.50, fetch_k=20, CrossEncoder reranking.
- Held-out result: 9/10 cases passed; composite score 0.941.
- Average metrics: Faithfulness 0.993 | Recall 0.977 | Precision 0.808 | Relevancy 0.939.
- Main remaining issue: the Accessible King benchmark question is underspecified; a focused clarification experiment raised precision to 0.961.

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Business Problem Overview and Solution Approach

Problem

- Staff handle about 1,200 guest queries per week; roughly 40% require multiple policy documents.
- Manual lookup takes 8–12 minutes and creates a 15–20% risk of incomplete or incorrect answers.
- Property-specific fees, room features, cancellation rules, and accessibility guidance are easy to mix up.

Solution approach

- Extract policy PDFs with Docling; preserve headings and tables as Markdown.
- Use token-aware chunking, domain embeddings, FAISS retrieval, MMR, and CrossEncoder reranking.
- Generate concise answers with GPT-4o-mini using a strict retrieved-context-only prompt.

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Data Understanding

- Knowledge base: four selectable-text PDF policy guides.
- Coverage: room/property information; booking and cancellation; refunds and additional charges; conduct and accessibility.
- Source structure includes headings, policy sections, Markdown tables, and a six-property matrix.
- Properties represented: ASH-ATL, ASH-BOS, ASH-CHI, ASH-DEN, ASH-MIA, and ASH-SEA.
- OCR is disabled because the supplied PDFs contain selectable text.
- Chunks are enriched with source, section, property-code, and topic metadata for focused retrieval.

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RAG - Chunking - Embedding Model

- Embedding model: HarishMaths/Hotel-Policy-Embedding.
- Tokenizer maximum length: 128 tokens.
- Embeddings run on CPU with normalized vectors and progress rendering disabled.
- The same model is used when building and loading the FAISS index.
- Domain-specific embeddings improve matching for hotel-policy terms such as late checkout, pet fees, and room features.

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RAG - Chunking - Chunker

- Chunker: RecursiveCharacterTextSplitter.from_huggingface_tokenizer.
- Chunk size: 512 tokens; overlap: 64 tokens.
- Split priority: paragraph, line, heading, sentence, word, then character boundaries.
- Atomic property rows are added for property-matrix entries so a single property and fee can be retrieved precisely.
- Metadata is added after splitting to preserve section, topic, and property identity.

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RAG - Chunking - Chunk Documents

- Total chunks created: 43, including atomic property rows.

Example focused chunk

```
## Property-specific matrix | Property: ASH-MIA

ASH-MIA Brickell | Parking: Valet, $55 per night | Pets: $100 per stay; 2 pets max
```

- One-property rows reduce contamination from unrelated properties and broad policy sections.
- Each chunk carries source, section, property_codes, and topics metadata.

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RAG - Vector Database

- Vector database: FAISS with 43 normalized document vectors.
- Index is persisted for the backend under backend/faiss_index.
- Final retrieval setting: k=5 chunks.
- Candidate search uses MMR lambda=0.50 and fetch_k=20.
- CrossEncoder ms-marco-MiniLM-L-6-v2 reranks candidates; topic and property metadata provide additional boosts.

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RAG - Vector Database - Retrieval Function

Query: What are the parking fees at the Atlanta and Miami hotels?

•Property routing identifies ASH-ATL and ASH-MIA; topic routing identifies parking.
•Candidate chunks are retrieved with MMR, then reranked for direct relevance.

Relevant evidence

ASH-ATL Midtown: self parking, $28 per night.

ASH-MIA Brickell: valet, $55 per night.

•The final context contains the two requested property rows instead of unrelated policy sections.

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RAG - Generation System - Load the LLM

- Generator: OpenAI-compatible ChatOpenAI wrapper.
- Model: gpt-4o-mini.
- temperature=0.0 | top_p=1.0 | max_tokens=512.
- API key and base URL are loaded from environment/configuration rather than hard-coded.
- Deterministic settings favor repeatable, concise policy answers for evaluation and operations.

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RAG - Generation System - Generation Prompt

Prompt inputs: user query + retrieved policy information.

- Use only information explicitly supported by retrieved policy text.
- Include relevant conditions, exclusions, limits, exceptions, and coverage details.
- Focus on the question; omit unrelated content and do not invent facts.
- If evidence is insufficient, return: The available policy documents do not contain enough information to answer this question.
- Additional guards focus Advance Purchase, Accessible King, and late-checkout questions on their relevant evidence.

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RAG - Generation System - RAG Function

Pipeline: query -> retrieve top-k chunks -> generate grounded answer -> return answer and evidence.

Example query: What is the cancellation policy for a flexible reservation?

Example answer: Flexible rates can normally be canceled by 6:00 PM local property time one day before arrival without a room charge. Later cancellation or no-show may be charged one night plus tax; the confirmation controls if it differs.

- The response endpoint can return the answer alone or the answer plus relevant chunks for verification.

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Evaluation - Load the Evaluation Data

- Development evaluation file: data/evaluation.csv.
- 10 records with question, expected answer, and ground-truth relevant context.

Sample question: What time can I check in at ASH-CHI Loop, and when do I need to be out?

Expected answer: Standard check-in is 3:00 PM and standard check-out is 12:00 PM local property time; early arrival and late departure depend on availability and approval.

- Multi-context records use <> separators so relevant policy sections can be evaluated together.

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Evaluation - Define the Evaluation Model

- Judge model: gpt-4o through the DeepEval OpenAIModel wrapper.
- The generator remains gpt-4o-mini with temperature=0.0, top_p=1.0, and max_tokens=512.
- The stronger judge model scores factual support, retrieval coverage, retrieval focus, and answer relevance.
- Evaluations run synchronously so each test case has a stable, auditable metric result.

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Evaluation - Define the Evaluation Metrics

- Faithfulness >= 0.85: every important answer claim is supported by retrieved context.
- Contextual Recall >= 0.80: retrieved context contains information needed for the expected answer.
- Contextual Precision >= 0.75: relevant context is prioritized over unrelated content.
- Answer Relevancy >= 0.75: answer directly addresses the user question.
- Composite = 0.35F + 0.25Recall + 0.20Precision + 0.20Relevancy.
- A case passes only when all four metric thresholds are met.

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Evaluation - Build and Run the Evaluation

- Retrieval tuning matrix selected chunk_size=512, overlap=64, MMR lambda=0.50, and fetch_k=20.
- Dev proxy coverage: 0.836; held-out test proxy coverage: 0.850.
- Full DeepEval is the acceptance test; proxy coverage is used only to narrow candidates.
- The evaluation creates LLMTestCase objects with input, expected output, actual output, retrieval context, and reference context.
- Cases are classified as AUTO ANSWER, ASSISTED REVIEW, or MANUAL LOOKUP from the composite score.
- Result: baseline k=5 is the selected configuration by highest composite score.

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Evaluation - Load the Test Data

- Held-out file: data/test.xlsx.
- 10 unseen questions covering cancellation, discounts, room occupancy, accessibility, parking, breakfast, rate changes, checkout, pets, and incidental holds.
  Sample question: How many people can stay in a Double Queen room?

  Expected answer: Up to 4 guests, subject to fire-code and property occupancy limits; recheck the room and rate if the group size changes.
- The held-out set is used only for final comparison and selection of the baseline.

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Evaluation - Build and Run the Evaluation

Held-out DeepEval comparison (10 cases)

Baseline k=5 F=.993 R=.977 P=.808 AR=.939 Composite=.941 Pass=9/10

Hybrid k=5 F=.995 R=.979 P=.787 AR=.922 Composite=.935 Pass=8/10

Hybrid k=6 F=.993 R=.982 P=.781 AR=.929 Composite=.935 Pass=8/10

Selected: baseline k=5 because it has the highest weighted composite score.

Only remaining failure: Accessible King room inclusions; Contextual Precision=.451.

The benchmark question is underspecified. A focused clarification plus single-chunk retrieval scored Precision=.961 and passed all four metrics.

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